We cannot find Kähler-Einstein metrics on all compact Kähler M:

if $Ric(\omega) = \lambda\omega$ then we must have

$$c_1(M) = \frac{\lambda}{2\pi}[\omega]$$

<u>Definition</u> A cohomology class $c \in H^2(M, \mathbb{R})$ is a <u>Kähler class</u> if M admits a Kähler metric ω such that $c = [\omega]$.

We cannot find KE metrics unless one of the following holds:

1) $c_1(M) = 0$. we can try to solve $Ric(\omega) = 0$ with $[\omega]$ any Kähler class

2) $c_1(M)$ is a Kähler class (M is Fano) can try to solve $Ric(\omega) = \omega$ with $[\omega] = 2\pi c_1(M)$

3) $-c_1(M)$ is a Kähler class can try to solve $Ric(\omega) = -\omega$, $[\omega] = -2\pi c_1(M)$

E.g. $\quad Ric(\omega_{FS}) = (n+1)\,\omega_{FS}$ on $\mathbb{CP}^n$

so $\quad c_1(\mathbb{CP}^n) = \frac{n+1}{2\pi}[\omega_{FS}]$

In chart U_0 using coords $w^i = \frac{z_i}{z_0}$, $i=1,\dots,n$

$$\omega_{FS} = \sqrt{-1}\,\partial\bar\partial \log(1+|w|^2)$$

has components

$$g_{j\bar k} = \partial_j \partial_{\bar k} \log(1+|w^1|^2+\dots+|w^n|^2)$$

$$= \partial_j \frac{w^k}{1+|w^1|^2+\dots+|w^n|^2} = \frac{\delta_{jk}}{1+|w|^2} - \frac{w^k \bar w^j}{(1+|w|^2)^2}$$

$$= \frac{1}{1+|w|^2}\left(\delta_{jk} - \frac{w^k \bar w^j}{1+|w|^2}\right)$$

$$\omega = \sqrt{-1}\sum g_{j\bar k}\, dz^j \wedge d\bar z^k$$

for rank 1 matrix A $\quad \det(I+A) = 1 + Tr(A)$

$$\Rightarrow \det(g_{j\bar k}) = (1+|w|^2)^{-n}\left(1 - \frac{|w|^2}{1+|w|^2}\right) = (1+|w|^2)^{-(n+1)}$$

$$Ric(\omega_{FS}) = -\sqrt{-1}\,\partial\bar\partial \log\det(g_{j\bar k}) = (n+1)\sqrt{-1}\,\partial\bar\partial\log(1+|w|^2)$$

$$= (n+1)\,\omega_{FS}$$

<u>E.g.</u> If $V \subset \mathbb{CP}^n$ is a degree d hypersurface

$$\left(\begin{array}{l} V = \{ [z_0 : \dots : z_n] \mid f(z_0, \dots, z_n) = 0 \} \\ \text{for homogeneous deg } d \text{ poly. } f \end{array} \right)$$

then $c_1(V) = \frac{1}{2\pi}(n+1-d)\,[\omega_{FS}]|_V$

$c_1(V) = 0$ if $d = n+1$

$c_1(V)$ is Kähler class if $d < n+1$

$-c_1(V)$ is Kähler class if $d > n+1$

<u>Theorem</u> (Yau) Let (M, ω) be compact Kähler and let α be a real $(1,1)$-form such that $\frac{1}{2\pi}[\alpha] = c_1(M)$. Then there is a unique Kähler metric η on M with $[\eta] = [\omega]$ and $Ric(\eta) = \alpha$.

<u>Reduction to PDE:</u>

By $\partial\bar{\partial}$-lemma $\quad Ric(\omega) = \alpha + \sqrt{-1}\,\partial\bar{\partial} F$

$F : M \to \mathbb{R}$

Suppose we can ~~solve~~ find $f : M \to \mathbb{R}$ such that $\eta = \omega + \sqrt{-1}\,\partial\bar{\partial} f$ is a Kähler form

and
$$\frac{\det(\omega + \sqrt{-1}\,\partial\bar{\partial} f)}{\det(\omega)} = e^F \qquad (*)$$

Then
$$-Ric(\eta) + Ric(\omega) = \sqrt{-1}\,\partial\bar{\partial} F$$
$$= Ric(\omega) - \alpha$$

so $\quad Ric(\eta) = \alpha$

Uniqueness follows from $\partial\bar{\partial}$-lemma and uniqueness of solutions of $(*)$ up to adding a constant.

Solution strategy (continuity method)

Instead of $(*)$ consider family of equations

$$\frac{\det(\omega + \sqrt{-1}\,\partial\bar\partial f)}{\det(\omega)} = e^{tF} \qquad (*_t)$$

s.t. $\omega + \sqrt{-1}\,\partial\bar\partial f$ are Kähler

for $t \in [0,1]$

Let $S \subset [0,1]$ be set of t for which $(*_t)$ admits a solution f_t.

Show 1. S is non-empty ($0 \in S$ since $f_0 = 0$ solves $(*_0)$)

2. S is open – use implicit function theorem in Banach spaces

3. S is closed, i.e. if f_{t_i} solves $(*_{t_i})$ and $t_i \to t_\infty$, we need to extract limit $f_{t_i} \to f_{t_\infty}$ solving $(*_{t_\infty})$

Uses apriori estimates

(need to normalize f_{t_i} by adding suitable constants)